**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI**

**COLLEGE OF ENGINEERING**

B.Sc. (Engineering), Mid Semester Examination, April, 2022
Third Year

***TE 385: SIGNALS AND SYSTEMS*** ***TIME: 1 HOUR***

Program of study: ..... INDEX NO: .....

**Attempt all questions in each section**
**Answer section A on the question paper and section B in the answer booklet**

**SECTION A (10 Marks)**

1 Continuous-time signals are defined for ...continuous values of independent variables.....

2 Discrete-time signals are .....discrete..... in time and

.....continuous..... in amplitude

3 For an even signal $x(-t) =$ ...x(t).....

4 $u(t-a) = 0$, if .....t<a.....

5 What are the two basic properties that Linear-Time Invariant Systems possess?
linearity and time-invariance
.....

6 The continuous time unit impulse signal is the .....first derivative..... of the continuous time unit step signal

7 A system which has a unique relation between its input and output is called

.....inverse system.....

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8 For a time-invariant system its ..... input and output relationship .....

..... do not change with time

9 If $x(t)$ is replaced by $x(at)$, then $x(at)$ is ..... linearly stretched .....
version of $x(t)$ if $|a| < 1$

10 All memoryless systems are causal systems. True/False ..... true .....

**SECTION B (50 Marks)**

1 (a) If $y[n] = x[n] * h[n]$, show that $\sum_y = \sum_x \sum_h$ where $\sum_x = \sum_{n=-\infty}^{\infty} x[n]$ and $\sum_h = \sum_{n=-\infty}^{\infty} h[n]$.

(b) Compute $y[n] = x[n] * h[n]$ when x[n] = {1, 2, 2} and h[n] = x[n] and check the correctness by using (a) above.

2 Sketch the signals $u(-t+1)$, $u(t-4)$ and $u[n] - u[n-1]$

3 What is a periodic signal? Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x(t) = 5\cos(7t + \frac{\pi}{6})$

(b) $x[n] = 7\cos(5\pi n + \frac{\pi}{6})$

(c) $x[n] = e^{j7\pi n}$

4 Consider the system $y[n] = sin\ x[n]$, state whether the system is

(a) Memoryless
(b) Linear
(c) Time invariant
(d) Causal

Justify your answers

5 Compute the convolution of x(t) = u(t) and h(t) = u(t)

**K. A.P. AGYEKUM**

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